

Ariq Koh Boon Xiong

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EDUCATION

Nanyang Technological University, Singapore

Aug 2024 – May 2028

Bachelor of Computing (Computer Science)

- Honours (Highest Distinction), Current CGPA: 4.68/5.00
- **Scholarship:** Turing AI Scholars Programme (TAISP) Scholarship
- **Relevant Coursework:** Artificial Intelligence (A+), Data Analytics & Mining (A+), Data Structures & Algorithms (A+), Digital Logic (A+), Machine Learning (A+), Object Oriented Design & Programming (A+), Probability (A+), Program Synthesis (A+), Algorithm Design & Analysis (A), Calculus (A), Computer Organisation & Architecture (A)

University of California, Berkeley, USA

Aug 2026 – Dec 2026

Berkeley Engineering GLOBE – Exchange Programme

PUBLICATIONS

QD-LLMs: Quality Diversity Optimization with LLMs for Generative Design Exploration

Jul 2026

First Author, In Proceedings – GECCO 2026 Workshop, San José, Costa Rica

Quality-Diversity LLM for Generative Design

Jun 2026

First Author, In Proceedings – AI4X Accelerate 2026, Singapore

EXPERIENCE

Makeable.build, San Francisco, California, USA

Jan 2027 – Aug 2027

AI and Software Engineering Intern

- Responsible for designing, experimenting, and deploying an autonomous pipeline for hardware manufacturing at an Afore Capital-backed startup.

A*STAR Centre for Frontier AI Research (CFAR), Singapore

Apr 2026 – Aug 2026

Research Intern – Agentic Scientist Platform

- Developed an Agentic Scientist Platform on a multi-agent framework for automated scientific research, implemented using Google ADK and incorporating modern agentic techniques including RAG, tool calling, and multi-agent orchestration.

Alibaba-NTU Global e-Sustainability CorpLab (ANGEL), Singapore

Jun 2025 – Dec 2025

Research Intern – Evolutionary Prompt Optimization

- Implemented a black-box automatic prompt optimization (APO) framework for LLM powered chatbot systems.
- Designed and built a multi-agent system built on evolutionary strategies to iteratively refine and optimize chatbot system prompts without access to model parameters.

PROJECTS

OneStop · iOS App with On-Device AI Assistant

Jan 2026 – Present

- Designed and deployed an all-in-one campus assistant app on the App Store, consolidating timetable management, bus tracking, and campus navigation into a single platform for NTU students.
- Built on-device AI chat powered by Apple Foundation Models, with automatic timetable extraction from screenshots and chatbot-integrated timetable queries and class reminders.
- Achieved a top 50 placing on the Singapore App Store Utilities chart, with over 700 unique downloads.

NTU Undergraduate Research Experience on Campus (URECA)

Aug 2025 – Jun 2026

- Proposed a novel framework for generative design exploration with LLM-based variation operators.
- Demonstrated more than 80% better physical performance in producing domain-aligned geometries against baseline methods, including iterative zero-shot prompting and Covariance Matrix Adaptation MAP-Elites (CMA-ME).

AWARDS

NTU Deep Learning Week 2026 · Finalist, Top 10 in Track

Mar 2026

- Built ResNet, an end-to-end post-crisis resource allocation system that analyzes disaster social media posts to predict emergency resource needs and visualize them on an interactive map, as part of a team of 3.
- Combined multimodal deep learning (CLIP ViT-B/32 encoder, ViT damage severity classifier, MLP resource classifier) with DBSCAN spatial clustering, and LLM-powered crisis summaries in a React + Leaflet frontend.

TAISP Hackathon 2025 · 3rd Place

Dec 2025

- Built real-time ASL recognition backend: MediaPipe hand landmark detection → TensorFlow/Keras neural network classifier (A-Z letters) → FastAPI WebSocket server streaming predictions at 30 FPS.
- Implemented GestureBuffer for temporal smoothing, pose analysis feedback, and word-completion detection mode for educational Visual Novel game integration.

SKILLS

AI-Assisted Development: Familiar with LLM coding agents (Claude Code, Codex) for prototyping and debugging

Languages: Python, C, C++, Java, Swift, English, Chinese

ML/AI: PyTorch, TensorFlow/Keras, pyribs (QD optimization), OpenAI API, LangChain, MediaPipe

Tools: Git, NumPy, OpenCV, SwiftUI, Node.js/Express